

SOFTWARE ENGINEERING

CASE STUDY REPORT

Case Study 36

SDG 9: Industry, Innovation and Infrastructure

Smart Construction Project Management System

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INTRODUCTION

1.1 BACKGROUND OF THE PROBLEM

Construction companies managing multiple projects simultaneously face significant operational challenges. Traditional methods relying on spreadsheets, emails, and manual reporting create fragmented information flows. Delays in material procurement, miscommunication among teams, and lack of real-time progress tracking frequently result in cost overruns, missed deadlines, and quality compromises.

1.2 IMPORTANCE OF SOLVING THE PROBLEM

The construction industry contributes substantially to economic development and employment. Inefficient project management leads to:

- Financial losses due to delays and rework
- Resource wastage including materials, labor, and equipment
- Safety risks from poor coordination
- Client dissatisfaction affecting business reputation

A centralized software solution can transform how construction projects are planned, executed, and monitored.

1.3 RELATED SDG GOAL(S)

SDG 9: Industry, Innovation, and Infrastructure — Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation.

This case study directly supports SDG 9 by introducing digital innovation to improve infrastructure development processes.

1.4 OBJECTIVES OF THE CASE STUDY

- Analyze challenges in current construction project management practices
- Propose a software engineering solution addressing identified gaps
- Design a system architecture with functional and non-functional requirements
- Recommend a suitable SDLC methodology for implementation

PROBLEM UNDERSTANDING

2.1 PROBLEM DESCRIPTION

Construction companies manage multiple concurrent projects using disconnected tools such as spreadsheets, paper-based reports, and informal communication channels. This fragmented approach creates several operational problems:

- **Material procurement delays** — No centralized tracking of inventory or supplier orders
- **Poor communication** — Information silos between site managers, engineers, and office staff
- **Lack of progress visibility** — Management cannot monitor real-time project status
- **Cost overruns** — Untracked expenses and resource misallocation
- **Document management issues** — Blueprints, contracts, and approvals scattered across systems

These inefficiencies compound across multiple projects, making timely decision-making nearly impossible.

2.2 OBJECTIVES

The proposed software solution aims to:

1. Centralize project planning, scheduling, and task management
2. Enable real-time progress tracking with visual dashboards
3. Automate material procurement and inventory management
4. Facilitate seamless communication among all stakeholders
5. Provide budget tracking and financial reporting
6. Maintain a centralized document repository
7. Generate analytical reports for decision-making
8. Ensure mobile accessibility for on-site personnel

2.3 SCOPE

In Scope:

- Project planning and scheduling module
- Task assignment and tracking
- Resource and material management
- Budget and expense tracking
- Document management
- Communication and notification system
- Reporting and analytics dashboard
- User role management (Admin, Project Manager, Site Engineer, Worker)
- Mobile application for field access

Out of Scope:

- Integration with external ERP systems (Phase 2)
- AI-based predictive analytics (Future enhancement)
- IoT sensor integration for equipment tracking
- Payroll and HR management

3. STAKEHOLDER ANALYSIS

Stakeholder	Role	Interest / Concern
Company Management	Strategic oversight	Real-time visibility into all projects, cost control, profitability
Project Managers	Day-to-day project coordination	Task scheduling, resource allocation, deadline adherence
Site Engineers	On-site technical supervision	Access to blueprints, daily progress updates, issue reporting
Site Workers	Task execution	Clear task assignments, safety information, reporting completion
Procurement Team	Material acquisition	Inventory tracking, supplier management, purchase orders
Finance Department	Budget management	Expense tracking, invoice processing, financial reports
Clients	Project sponsors	Progress visibility, milestone tracking, quality assurance
Suppliers / Vendors	Material providers	Order notifications, delivery schedules, payment tracking
IT Department	System maintenance	System reliability, security, user support

EXISTING SYSTEM ANALYSIS

4.1 EXISTING PROCESS

The current construction project management process operates as follows:

1. **Project initiation** — Contracts and plans stored in physical files or shared drives
2. **Task assignment** — Communicated verbally or via email/WhatsApp
3. **Progress tracking** — Site supervisors submit weekly paper reports
4. **Material management** — Procurement requests made via phone calls or emails
5. **Budget tracking** — Finance team maintains separate spreadsheets
6. **Communication** — Informal channels with no audit trail
7. **Reporting** — Manual compilation of data for management reviews

4.2 STRENGTHS

- Familiar to existing staff
- Low initial technology investment
- Flexibility in informal processes

4.3 WEAKNESS

- **Data fragmentation** — Information scattered across multiple formats and locations
- **Delayed information** — Weekly reports provide outdated snapshots
- **Error-prone** — Manual data entry leads to mistakes
- **Poor traceability** — No audit trail for decisions and changes
- **Limited scalability** — Processes break down with multiple concurrent projects
- **Communication gaps** — Critical information lost in informal channels
- **Inconsistent reporting** — No standardized formats or metrics

4.4 GAP ANALYSIS

Current State	Desired State	Gap
Manual task tracking	Automated task management with notifications	No digital workflow
Weekly progress reports	Real-time progress dashboards	Delayed visibility
Scattered documents	Centralized document repository	No single source of truth
Informal procurement	Systematic inventory and order management	No procurement workflow
Spreadsheet-based budgeting	Integrated financial tracking	Disconnected financial data
No mobile access	Mobile-enabled field operations	Limited accessibility

PROPOSED SOFTWARE SOLUTION

5.1 PROPOSED SYSTEM

The Smart Construction Project Management System (SCPMS) is a web and mobile-based platform that centralizes all aspects of construction project management. The system provides:

- A unified dashboard for managing multiple projects
- Real-time progress tracking with milestone visualization
- Integrated resource, material, and budget management
- Secure document storage with version control
- Role-based access for different stakeholder types
- Mobile application for on-site personnel
- Automated notifications and alerts

5.2 FUNCTIONAL MODULES

- User Registration & Authentication
- Project Planning & Scheduling
- Task Assignment & Tracking
- Resource Management
- Material Procurement Management
- Equipment Management
- Progress Monitoring Dashboard
- Document Management
- Notification & Alert System
- Report & Analytics Module
- Client Monitoring Portal
- Admin Dashboard

5.3 FUNCTIONAL REQUIREMENTS

ID	Requirement
FR01	The system shall allow administrators to create and manage user accounts with role-based permissions
FR02	The system shall enable project managers to create new projects with defined timelines, budgets, and milestones
FR03	The system shall provide Gantt chart visualization for project scheduling
FR04	The system shall allow task creation, assignment to specific users, and deadline setting
FR05	The system shall enable site engineers to update task progress with photo/document attachments
FR06	The system shall track material inventory with low-stock alerts
FR07	The system shall generate and manage purchase orders with supplier tracking
FR08	The system shall record all project expenses and compare against allocated budgets
FR09	The system shall store documents with version history and access control
FR10	The system shall send automated notifications for task deadlines, approvals, and alerts
FR11	The system shall provide a mobile application for field access to tasks and reporting
FR12	The system shall generate customizable reports on project progress, finances, and resources

5.4 NON FUNCTIONAL REQUIREMENTS

ID	Category	Requirement
NFR01	Performance	Dashboard pages shall load within 3 seconds under normal load
NFR02	Scalability	System shall support up to 100 concurrent users and 50 active projects
NFR03	Availability	System uptime shall be 99.5% excluding scheduled maintenance
NFR04	Security	All data transmission shall use TLS 1.3 encryption
NFR05	Security	Role-based access control shall restrict data visibility per user role
NFR06	Usability	Interface shall be navigable with minimal training (< 2 hours)
NFR07	Reliability	System shall perform automatic daily backups with 30-day retention
NFR08	Portability	Mobile application shall support Android 10+ and iOS 14+
NFR09	Maintainability	Codebase shall follow modular architecture for independent module updates
NFR10	Compliance	System shall maintain audit logs for all user actions for 1 year

5.5 SOFTWARE ENGINEERING PRINCIPLES APPLIED

Principle	Application
Modularity	System divided into independent modules (Projects, Tasks, Procurement, Finance) that can be developed and maintained separately
Scalability	Cloud-based architecture allows horizontal scaling as user base and project count grows
Reliability	Automated backups, error handling, and failover mechanisms ensure consistent operation
Security	Encryption, role-based access, and audit logging protect sensitive project and financial data
Maintainability	Clean code architecture with documentation enables future enhancements
Reusability	Common components (notifications, file upload, reporting) shared across modules
Usability	Intuitive interface design with consistent navigation patterns
Flexibility	Configurable workflows and custom fields accommodate different project types

6.1 SYSTEM DIAGRAM

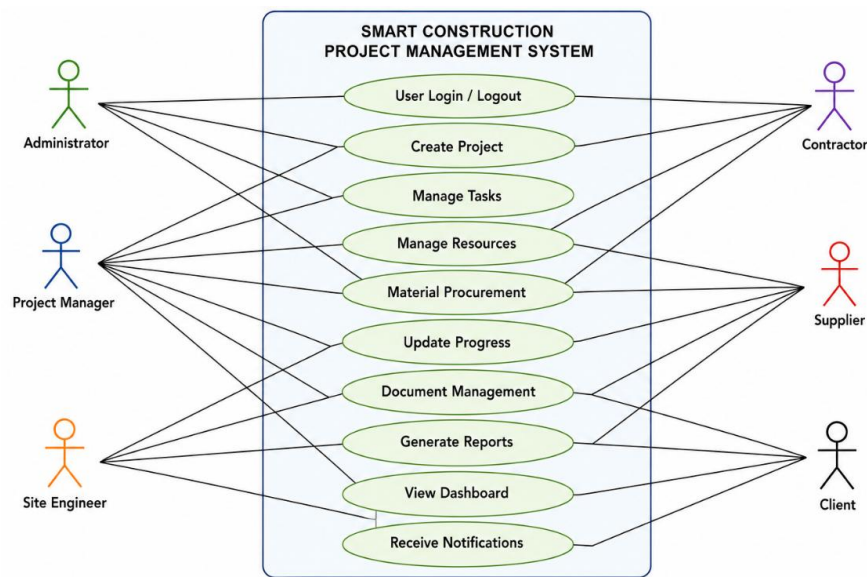


Figure 1: Use Case Diagram showing interactions between the Administrator, Project Manager, Site Engineer, Contractor, Supplier, Client, and the Smart Construction Project Management System.

6.2 ACTIVITY DIAGRAM-TASK MANAGEMENT FLOW

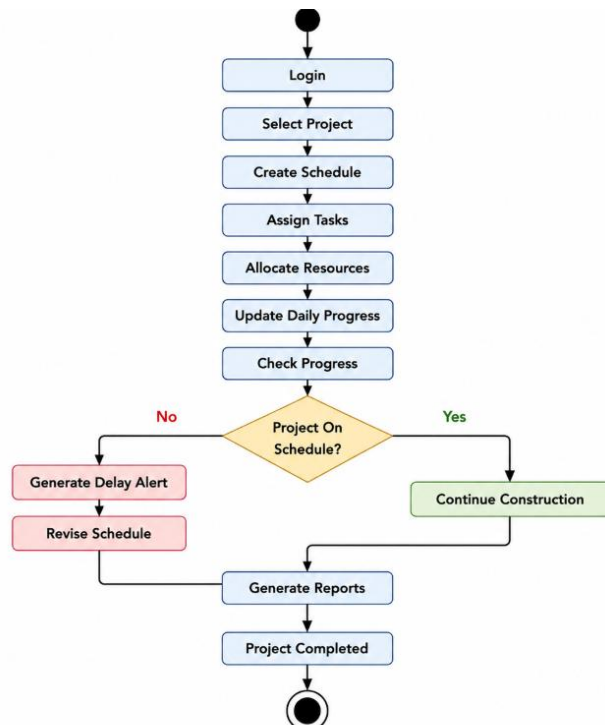


Figure 2: Activity Diagram illustrating the workflow of project planning, monitoring, and completion.

6.3 SYSTEM ARCHITECTURE DIAGRAM

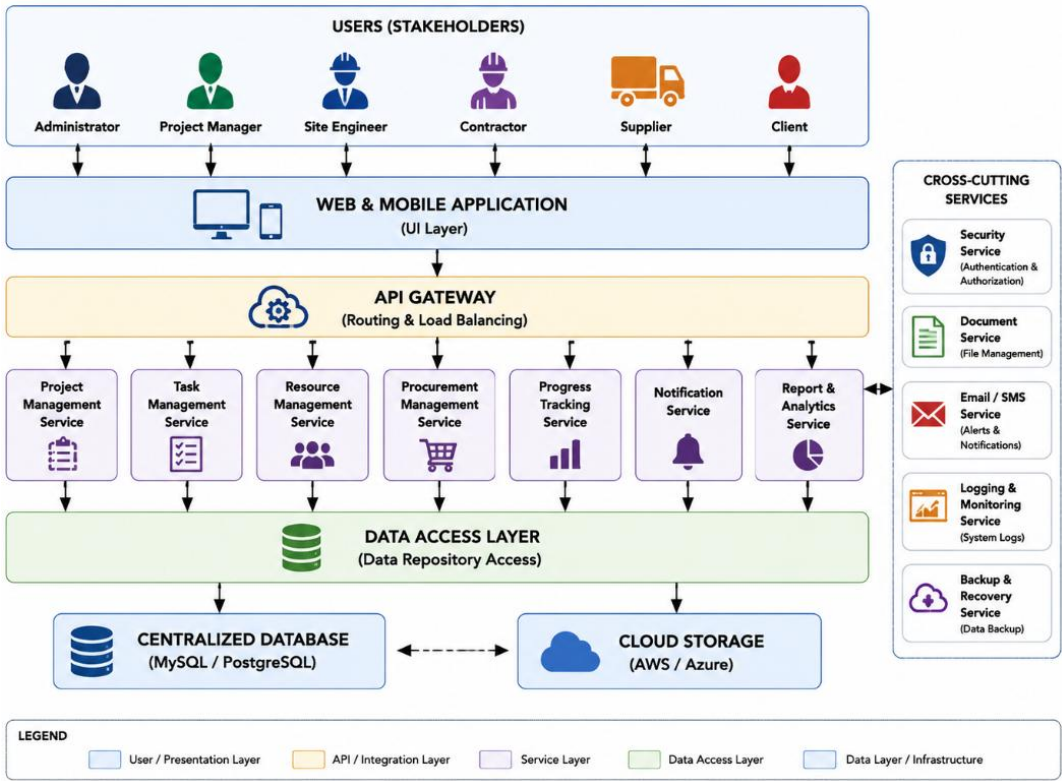


Figure 3: Layered architecture showing user applications, backend services, and the centralized cloud database.

6.4 DATABASE ER DIAGRAM

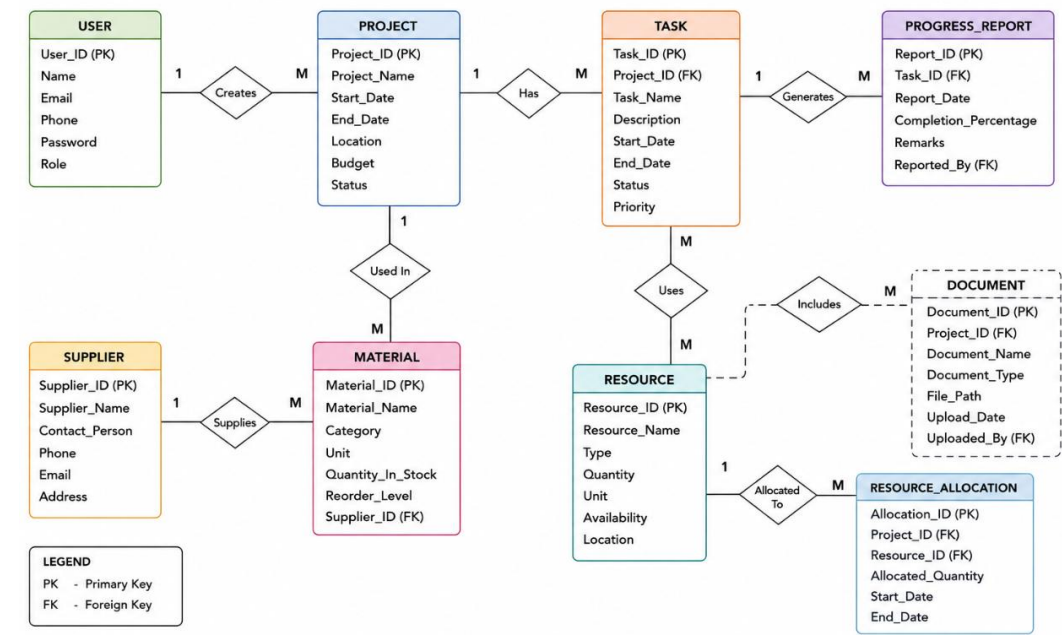


Figure 4: ER Diagram showing the relationships among Users, Projects, Tasks, Resources, Suppliers, Materials, and Progress Reports.

6.5 WORKFLOW DIAGRAM

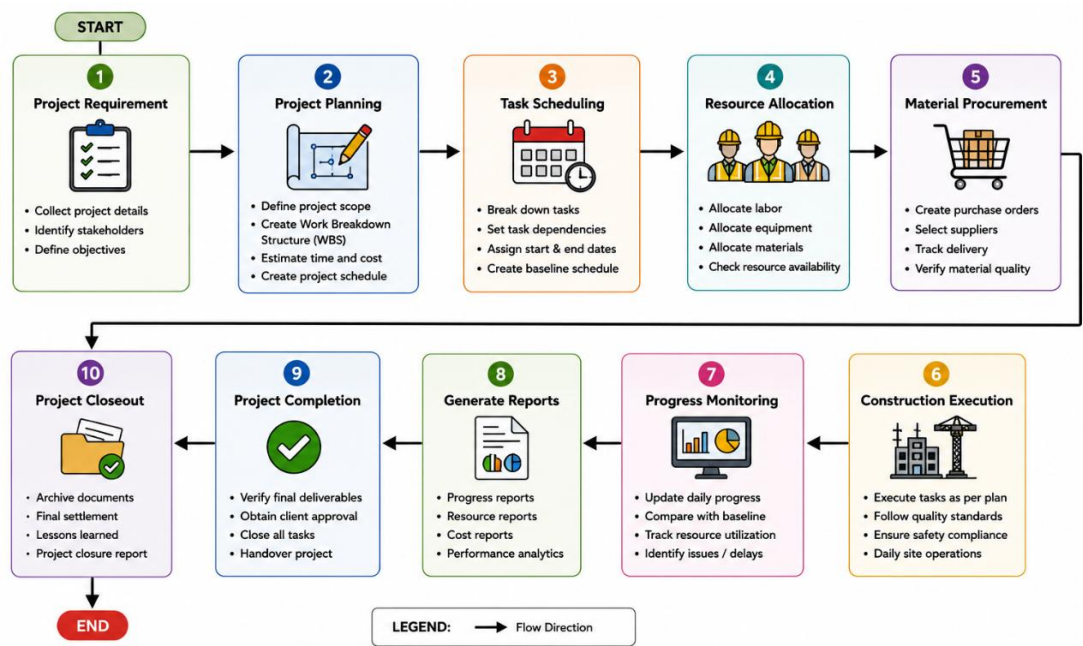


Figure 5: Workflow of the Smart Construction Project Management System from planning to project completion.

6.6 WIREFRAME/UI SCREENS

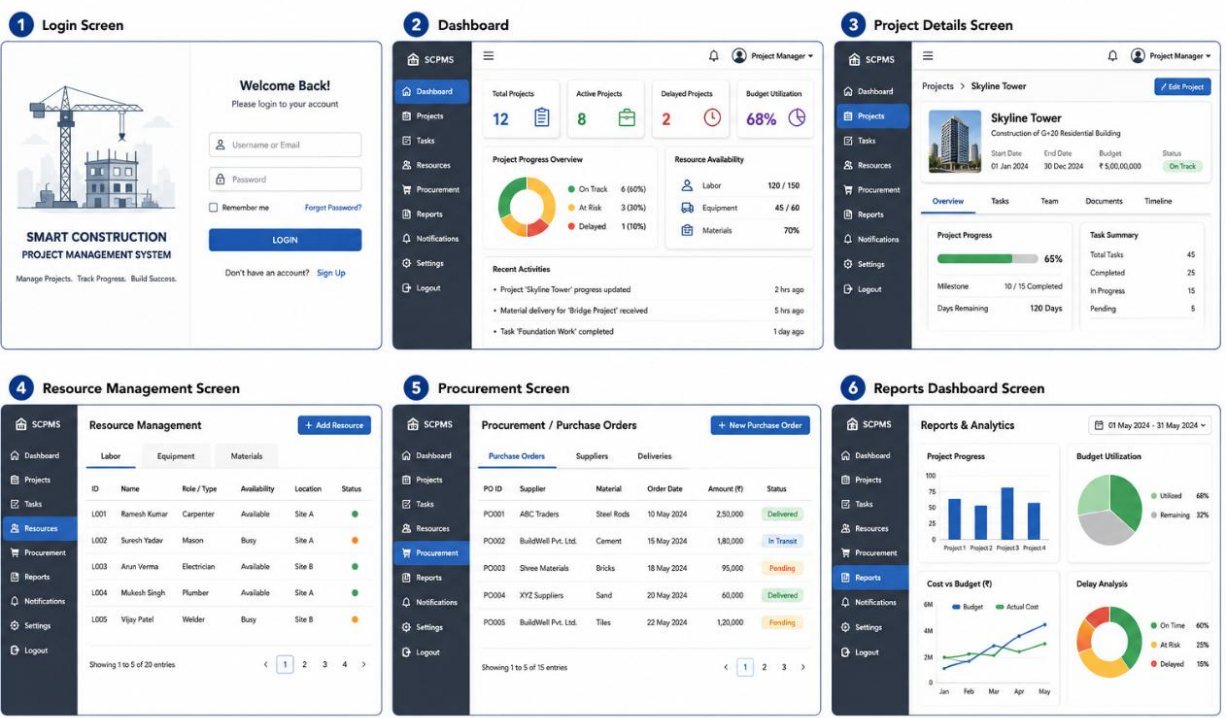


Figure 6: Prototype wireframes showing the Login page, Dashboard, Project Details, Resource Management, Procurement, and Reports screens.

7. METHODOLOGY/SDLC SCRUM

7.1 SELECTED MODEL

The Agile (Scrum) Model is selected for developing the Smart Construction Project Management System.

7.2 WHY AGILE(SCRUM)?

Construction project requirements frequently change due to client requests, design modifications, weather conditions, and resource availability. Agile allows the development team to deliver software in small iterations (sprints), receive stakeholder feedback, and continuously improve the system. This approach minimizes development risks and ensures that the software meets user requirements effectively.

7.3 ADVANTAGES OF AGILE FOR THIS PROJECT

- Faster software delivery.
- Continuous stakeholder feedback.
- Easy adaptation to changing requirements.
- Better collaboration among developers and users.
- Improved software quality through continuous testing.
- Reduced project risk.
- Incremental feature development.

7.4 SUITABILITY

Agile is suitable because different modules such as project planning, procurement, resource management, progress tracking, and reporting can be developed independently and integrated gradually, allowing users to evaluate each module before the complete system is deployed.

7.5 DEVELOPMENT STAGES(SPRINTS)

Sprint	Focus Area	Deliverables
Sprint 0	Requirement Analysis	SRS, Planning
Sprint 1	User Management	Login, Registration
Sprint 2	Project & Task Module	Project Creation, Scheduling
Sprint 3	Resource & Procurement	Material and Resource Management
Sprint 4	Progress Monitoring	Dashboard, Reports
Sprint 5	Testing & Deployment	Bug Fixes, Final Deployment

8.IMPLEMENTATION PLAN

Phase	Activities	Duration	Deliverables
Phase 1: Planning	Requirements analysis, stakeholder interviews, architecture design	Weeks 1-2	SRS document, System design
Phase 2: Core Development	User management, project module, dashboard	Weeks 3-8	MVP with basic functionality
Phase 3: Task & Mobile	Task management, progress tracking, mobile app	Weeks 9-14	Field-ready mobile application
Phase 4: Operations	Procurement, inventory, finance modules	Weeks 15-22	Complete operational modules

Phase	Activities	Duration	Deliverables
Phase 5: Reporting	Document management, analytics, reports	Weeks 23-26	Full reporting capabilities
Phase 6: Testing	Integration testing, UAT, performance testing	Weeks 27-28	Test reports, bug fixes
Phase 7: Deployment	Production deployment, training, go-live	Weeks 29-30	Live system, trained users

9.EXPECTED OUTCOMES

Benefits

- Faster project planning and scheduling.
- Real-time project progress monitoring.
- Improved communication among stakeholders.
- Better resource and material utilization.
- Reduced project delays and cost overruns.
- Accurate reporting and project analytics.
- Improved customer satisfaction.

Business Impact

- Increased productivity.
- Reduced operational costs.
- Better decision-making.
- Improved profitability.
- Efficient project execution.

SDG Contribution

The proposed system supports SDG 9 – Industry, Innovation and Infrastructure by promoting digital transformation, smart construction management, and sustainable infrastructure development.

10.RISKS AND LIMITATIONS

Internet Dependency

Cloud-based services require reliable internet connectivity.

Data Security

Sensitive project documents must be protected from unauthorized access.

User Resistance

Employees may require training before adopting the new system.

Budget Constraints

Initial software development and deployment costs may be high.

System Downtime

Unexpected server failures may temporarily interrupt project monitoring.

Data Accuracy

Incorrect data entered by users may affect project reports and decision-making.

11.FUTURE ENHANCEMENTS

- Artificial Intelligence for project delay prediction.
- Machine Learning for resource optimization.
- IoT integration for real-time equipment monitoring.
- Drone-based construction site inspection.
- BIM (Building Information Modeling) integration.
- Mobile application with offline synchronization.
- Predictive analytics dashboards.
- Voice-enabled project assistant.
- Cloud-based multi-company support.

12. CONCLUSION

The Smart Construction Project Management System provides an efficient software engineering solution for managing construction projects through centralized planning, resource management, procurement tracking, and real-time progress monitoring. The proposed cloud-based system improves communication among stakeholders, reduces project delays, minimizes operational costs, and enhances transparency.

Using the Agile (Scrum) methodology enables incremental development, continuous stakeholder feedback, and flexible adaptation to changing project requirements. The proposed solution supports Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) by encouraging the use of digital technologies to improve construction project management, resulting in more efficient, innovative, and sustainable infrastructure development.

13. REFERENCES

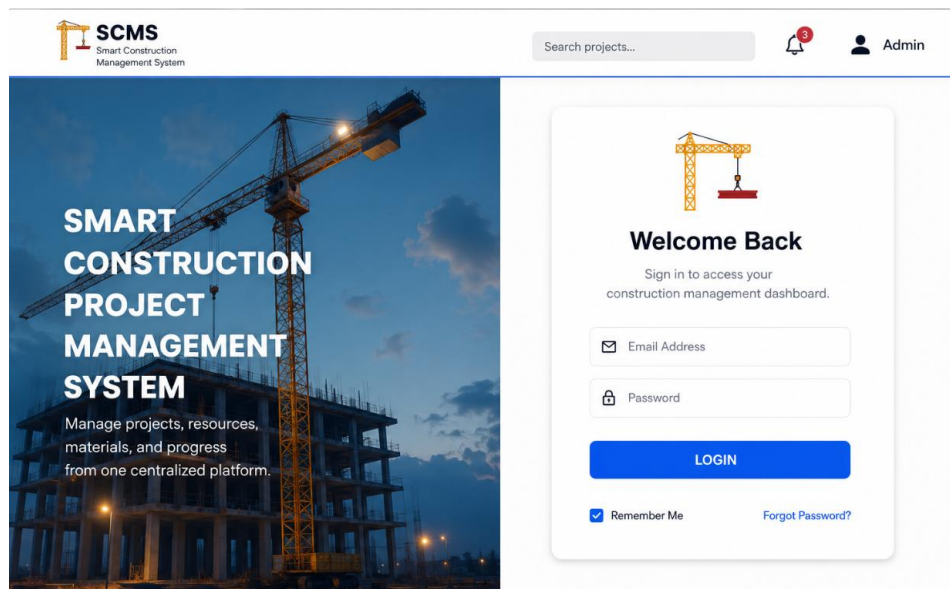
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14. APPENDIX

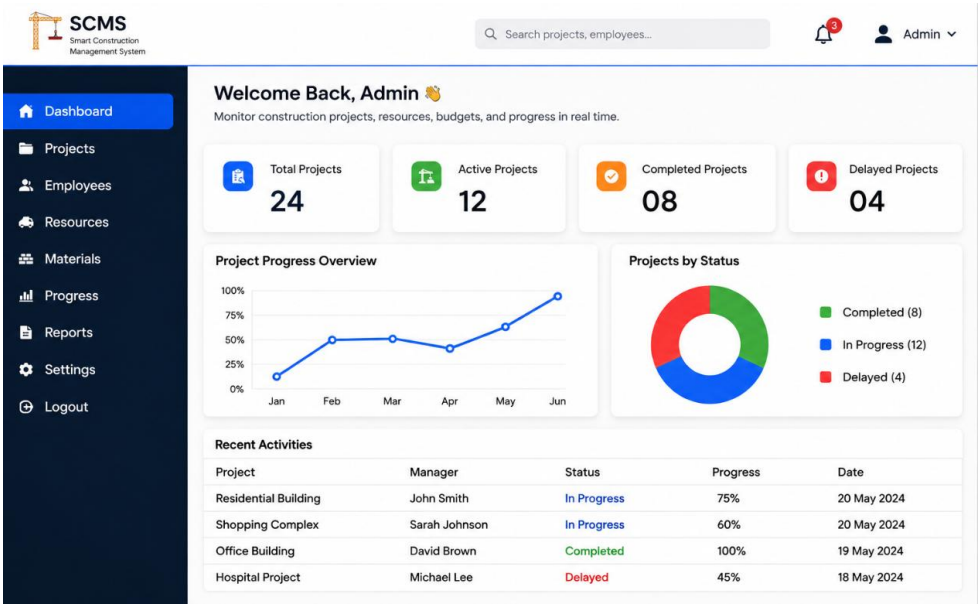
The appendix may include:

- Software Requirement Specification (SRS)
- Use Case Diagram
- Activity Diagram
- System Architecture Diagram
- ER Diagram
- Workflow Diagram
- UI Wireframes (Figma/Canva)
- Sample Project Dashboard
- Sample Progress Reports
- Test Cases
- Screenshots of the developed prototype

1.LOGIN PAGE



2. Dashboard



3. Project Management

SCMS
Smart Construction
Management System

Search projects, employees...

3

Admin

Dashboard

Projects

Employees

Resources

Materials

Progress

Reports

Settings

Logout

Projects

+ Add Project

Project Name	Location	Manager	Start Date	End Date	Status	Progress	Action
Residential Building	New York	John Smith	01 Jan 2024	31 Dec 2024	In Progress	75%	...
Shopping Complex	Los Angeles	Sarah Johnson	15 Feb 2024	15 Dec 2024	In Progress	60%	...
Office Building	Chicago	David Brown	10 Mar 2024	10 Oct 2024	Completed	100%	...
Hospital Project	Houston	Michael Lee	20 Jan 2024	20 Nov 2024	Delayed	45%	...
Bridge Construction	Miami	James Wilson	05 Apr 2024	05 Jan 2025	In Progress	30%	...

4. Employees Management

SCMS
Smart Construction
Management System

Search projects, employees...

3

Admin ▾

Dashboard

Projects

Employees

Resources

Materials

Progress

Reports

Settings

Logout

Employees

+ Add Employee

Employee ID	Name	Position	Department	Email	Phone	Status
EMP001	John Smith	Project Manager	Management	john@gmail.com	1234567890	Active
EMP002	Sarah Johnson	Site Engineer	Engineering	sarah@gmail.com	1234567891	Active
EMP003	David Brown	Civil Engineer	Engineering	david@gmail.com	1234567892	Active
EMP004	Michael Lee	Safety Officer	Safety	michael@gmail.com	1234567893	Active
EMP005	James Wilson	Supervisor	Site	james@gmail.com	1234567894	Inactive

5. Resources Management

SCMS
Smart Construction
Management System

Search projects, employees...

3

Admin ▾

Dashboard

Projects

Employees

Resources

Materials

Progress

Reports

Settings

Logout

Resources

+ Add Resource

Resource Name	Type	Quantity	Unit	Status	Location	Actions
 Excavator	Equipment	5	Nos	Available	Site A	...
 Concrete Mixer	Equipment	3	Nos	In Use	Site B	...
 Tower Crane	Equipment	2	Nos	Available	Site A	...
 Generator	Equipment	4	Nos	Maintenance	Site C	...
 Skilled Workers	Manpower	25	Persons	Available	Site A	...
 Truck	Vehicle	6	Nos	In Use	Site B	...

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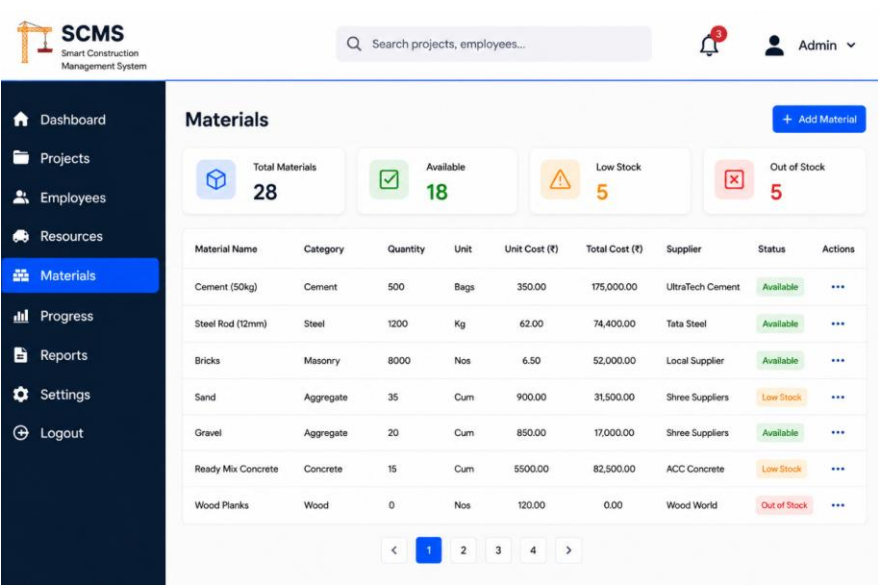
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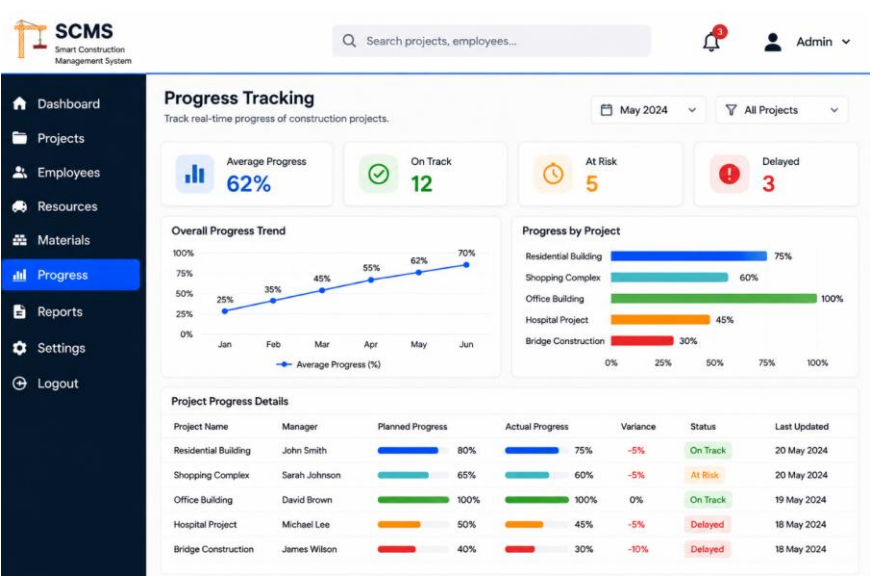
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6. Materials Management



7. Progress Tracking



8. Reports

